MAT 3103: Computational Statistics and Probability **Semester: Fall 2019-20**

Chapter 4: Discrete Probability Distribution

Ashfia Tasnim Munia, Md. Mortuza Ahmmed, Dr. Mahfuza Khatun.

Probability Distribution: Probability distribution is the distribution of random variable.

Types of probability distribution

There are two types of probability distribution.

1. Discrete probability distribution
   - Binomial
   - Poisson
2. Continuous probability distribution
   - Normal
   - Exponential
   - Rayleigh

4.1 Discrete probability distribution

4.1.1 Binomial distribution:

Binomial experimental results are of 2 types, denoted by 'Success' with probability 'p' and 'Failure' with probability 'q'. Assume that, out of n trials there are x success and $(n-x)$ failures. If X is a discrete random variable indicating the number of successes, then probability distribution of x successes is called Binomial distribution. It is given by

$$P(X = x) = \binom{n}{x} p^x q^{n-x}; n = 0, 1, 2, \ldots, n$$

Here, $p + q = 1$
Mean, $E(X) = np$ and Variance, $V(X) = npq$

Examples:

- Randomly selected good or defective products of an industry.
- Randomly selected literate or illiterate persons of a society.
- Randomly selected inured or non-injured industrial worker.
- Randomly selected machines operated successfully or not.

4.1.1.1 Mean and variance calculation using generating functions

For discrete random variable X, the moment generating function(mgf) is,

$$\begin{aligned} M_x(t) = E[e^{tX}] &= \sum e^{tX} P(X) \\ &= \sum_{x=0}^{n} e^{tX} \binom{n}{x} p^x q^{n-x} \\ &= q^n + \binom{n}{1}(pe^t)q^{n-1} + \binom{n}{2}(pe^t)^2 q^{n-2} + \ldots + (pe^t)^n \\ &= (q + pe^t)^n \end{aligned}$$

We know, $\mu'_r = \frac{d^r M_x(t)}{dt^r}|_{t=0}$

Similarly, we can calculate,

$$\begin{aligned} \mu'_1 &= \frac{dM_x(t)}{dt}|_{t=0} \\ &= \frac{d}{dt}[(q + pe^t)^n]|_{t=0} \\ &= n(q + pe^t)^{n-1}(0 + pe^t)|_{t=0} \\ &= n(q + p)^{n-1} p \\ &= np \end{aligned}$$

Mean, $E(X) = \mu'_1 = np$

$$\begin{aligned} \mu'_2 &= \frac{d^2 M_x(t)}{dt^2}|_{t=0} \\ &= \frac{d}{dt}[n(q + pe^t)^{n-1} pe^t]|_{t=0} \\ &= np\frac{d}{dt}[(q + pe^t)^{n-1} e^t]|_{t=0} \\ &= np[e^t(n-1)(q + pe^t)^{n-2} pe^t + (q + pe^t)^{n-1} e^t]|_{t=0} \\ &= np[(n-1)(q + p)^{n-2} p + (q + p)^{n-1}] \\ &= np[(n-1)p + 1] \\ &= n^2 p^2 - np^2 + np \end{aligned}$$

Variance, $V(X) = \mu_2$,

$$\begin{aligned}\mu_2 &= \mu_2' - \mu_1'^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= -np^2 + np \\ &= np(1-p) \\ &= npq\end{aligned}$$

So, Mean= np and Variance= npq.

4.1.1.2 Example of Binomial Distribution

Example 1: 80% devices of a workshop work properly. Once 10 devices are selected randomly. Find the probability that, out of 10, (i) all work properly, (ii) at best 2 works properly, (iii) at least 3 works properly, (iv) 2 to 4 work properly, (v)Find the mean and variance of the distribution.

Solution: Let, X be the number of properly worked devices. Given, $n = 10$, $p = 0.8$, and $q = 1p = 0.2$. So,

$$\begin{aligned}P(X = x) &= \binom{n}{x} p^x q^{n-x} \\ &= \binom{10}{x} (0.8)^x (0.2)^{10-x}; n = 0, 1, 2, \ldots, 10\end{aligned}$$

(i) $P(X = 10) = \binom{10}{10}(0.8)^{10}(0.2)^{10-10} = 0.1073$

(ii) $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = [\binom{10}{0}(0.8)^0(0.2)^{10-0}] + [\binom{10}{1}(0.8)^1(0.2)^{10-1}] + [\binom{10}{2}(0.8)^2(0.2)^{10-2}] = 0.000078$

(iii) $P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] = 1 - 0.000078 = 0.99992$

(iv) $P(2 \leq X \leq 4) = [P(X = 2) + P(X = 3) + P(X = 4)] = [\binom{10}{2}(0.8)^2(0.2)^{10-2}] + [\binom{10}{3}(0.8)^3(0.2)^{10-3}] + [\binom{10}{4}(0.8)^4(0.2)^{10-4}] = 0.006365$

(v) Mean=$np = 10 \times 0.8 = 8$ and Variance=$npq = 10 \times 0.8 \times 0.2 = 1.6$

Example 2: A test is conducted which is consisting of 20 MCQs with every MCQ having its 4 options out of which only 1 is correct. Determine the probability that a person undertaking that test has answered exactly 5 questions wrong.

Solution: Let, X be the number of wrong answers. Given, $p = \frac{3}{4} = 0.75, q = 1 - p = 1 - 0.75 = 0.25$ and $n = 20$.

$P(X = 5) = \binom{n}{x}p^x q^{n-x} = \binom{20}{5}(0.75)^5(0.25)^{20-5} = 0.0000034$

4.1.2 Poisson distribution:

It is a limiting case of the binomial distribution under the conditions:

(a) $n \to \infty (verylarge)$

(b) $p \to 0(verysmall)$

(c) $np = \lambda; \lambda > 0$

If X is a Poisson random variable, then its distribution is:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}; x \geq 0$$

Mean=Variance=λ.

Examples:

- Number of defective materials produced in an industry.
- Number of wrong connections received at a telephone exchange.
- Number of printing mistakes at each page of a book.
- Number of faded out signals sent from a station.

4.1.2.1 Mean, variance & shape characteristics calculation using generating functions

For discrete random variable X, the moment generating function(mgf) is,

$$\begin{aligned} M_x(t) = E[e^{tX}] &= \sum e^{tX} P(X) \\ &= \sum_{x=0}^{\infty} e^{tX} \frac{e^{-\lambda}\lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \\ &= e^{-\lambda}[1 + \lambda e^t + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \ldots] \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t - 1)} \end{aligned}$$

The Cumulant generating function is, $K_x(t)$

$K_x(t) = lnM_x(t) = lne^{\lambda(e^t-1)} = \lambda(e^t-1) = \lambda(1+t+\frac{t^2}{2!}+\frac{t^3}{3!}+\ldots-1) = \lambda(t+\frac{t^2}{2!}+\frac{t^3}{3!}+\ldots) = \lambda t+\lambda\frac{t^2}{2!}+\lambda\frac{t^3}{3!}+\ldots$

K_r = Coefficient of $\frac{t^r}{r!}$ in $K_x(t)$

$K_1 = \mu_1'$ =Mean=λ

$K_2 = \mu_2$ =Variance=λ

$K_3 = \mu_3 = \lambda$

$K_4 = \mu_4 - 3\mu_2^2$

So, $\mu_4 = K_4 + 3\mu_2^2 = \lambda + 3\lambda^2$

Skewness, $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\lambda^2}{\lambda^3} = \frac{1}{\lambda} > 0$, So the distribution is positively skewed or right skewed.

Kurtosis, $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\lambda+3\lambda^2}{\lambda^2} = \frac{1}{\lambda} + 3 > 3$, So the distribution is leptokurtic.

4.1.2.2 Example of Poisson Distribution

Example 1: The average number of signals sent from a station reaching properly to another station is 3 per day. Find the probability that the signals which are sent in a day reached properly are (i) 3, (ii) at best 2, (iii) at least 3.

Solution: Let, X be the number of not properly reached signals. Given, $\lambda = 3$.

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-3}3^x}{x!}; x \geq 0$$

(i) $P(X = 3) = \frac{e^{-3}3^3}{3!} = 0.22404$

(ii) $P(X \leq 2) = [P(X = 0) + P(X = 1) + P(X = 2)] = [\frac{e^{-3}3^0}{0!}] + [\frac{e^{-3}3^1}{1!}] + [\frac{e^{-3}3^2}{2!}] = 0.42319$

(iii) $P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] = 1 - 0.42319 = 0.57681$

Example 2: A company makes electric motors. The probability that an electric motor is defective is 0.01. What is the probability that a sample of 300 electric motors will contain exactly 5 defective motors?

Solution: Let, X be the no. of defective motors. Given, $n = 300$, $p = 0.01$. So, $\lambda = np = 300 \times 0.01 = 3$. Then

$P(X = 5) = \frac{e^{-3}3^x}{x!} = \frac{e^{-3}3^5}{5!} = 0.10082$.

4.2 Related Sample questions

1. Write down 2 examples of each of binomial variable and Poisson variable.
2. Under what conditions a binomial variable transform to Poisson variable?
3. Eighty percent computers of a laboratory are good. Once 8 computers are selected at random. Find the probability that, out of 8 computers; (i) at least 2 are good, (ii) 3 are good, (iii) at best 3 are good, (iv) 2 to 4 are good computers, (v) Find mean and variance of number of good computers.
4. Three percent computers of a company are found defective. The company produces 100 computers per day. Find the probability that in a randomly selected day- (i) at least 2, (ii) at best 3, (iii) 1 to 4 are defective computers.
5. The average number of defective circuits produced in an industry per day is 3. Find the probability that in a randomly selected day there will be- (i) at least 2, (ii) at best 3, (iii) 1 to 3 defective circuits.